

Linux Commands Handbook for Git Bash

Beginner to Intermediate Self-Study Guide

This handbook introduces essential Linux commands that work in Git Bash on Windows. Each section includes syntax, explanation, examples, and practice exercises.

1. Navigation Commands

Command	Description	Example
pwd	Print current working directory	pwd
ls	List files	ls -la
cd	Change directory	cd project
clear	Clear terminal	clear

Notes: Practice each command in Git Bash. Use `--help` where available to explore additional options.

2. File Operations

Command	Description	Example
touch	Create file	touch notes.txt
cat	Display file	cat notes.txt
cp	Copy file	cp a.txt b.txt
mv	Move/Rename	mv old.txt new.txt
rm	Delete file	rm temp.txt

Notes: Practice each command in Git Bash. Use `--help` where available to explore additional options.

3. Directory Operations

Command	Description	Example
mkdir	Create directory	mkdir src
mkdir -p	Create nested directories	mkdir -p a/b/c
rmdir	Remove empty directory	rmdir temp
rm -r	Delete directory recursively	rm -r build

Notes: Practice each command in Git Bash. Use `--help` where available to explore additional options.

4. Searching

Command	Description	Example
find	Find files	find . -name '*.java'
grep	Search text	grep 'main' Hello.java

Notes: Practice each command in Git Bash. Use --help where available to explore additional options.

5. Permissions

Command	Description	Example
chmod	Change permissions	chmod +x script.sh

Notes: Practice each command in Git Bash. Use --help where available to explore additional options.

6. Networking

Command	Description	Example
ping	Test connectivity	ping google.com
curl	Fetch URL	curl https://example.com
ssh	Remote login	ssh user@host

Notes: Practice each command in Git Bash. Use --help where available to explore additional options.

7. Environment

Command	Description	Example
echo	Display variable	echo \$PATH
export	Set variable	export JAVA_HOME=/c/java

Notes: Practice each command in Git Bash. Use --help where available to explore additional options.

8. History & Shortcuts

Command	Description	Example
history	Show history	history
Ctrl+R	Search history	Keyboard shortcut

Notes: Practice each command in Git Bash. Use --help where available to explore additional options.

9. Practice Workflow

Command	Description	Example
Project	Typical session	mkdir Demo && cd Demo && touch app.txt && ls -la

Notes: Practice each command in Git Bash. Use `--help` where available to explore additional options.

Mini Lab

1. Create a Training folder with Java and Python subfolders.
2. Create three text files using touch.
3. Move a file into a subfolder using mv.
4. Copy a file using cp.
5. Search for *.txt files using find.
6. Display hidden files using ls -la.
7. Search for text using grep.
8. Delete a directory recursively using rm -r.

Top 25 Commands to Memorize

pwd, ls, ls -la, cd, mkdir, touch, cat, less, head, tail, cp, mv, rm, find, grep, echo, history, clear, whoami, date, chmod, curl, ssh, export, tree.